SkillBridge AI

Bridging the Gap Between Education and Employment

Youth Employment and Work Readiness
FNB App of the Year Hackathon 2026

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A practical prototype for graduate career preparation

The Problem: From Education to Employment

The challenge

Graduates may struggle to translate their qualifications and existing skills into relevant entry-level employment opportunities.

01. Finding opportunities

Identifying relevant entry-level jobs can be difficult.

02. Skills gaps

Graduates may not know which skills a particular role requires.

03. CV preparation

Tailoring a CV to a job description can be challenging.

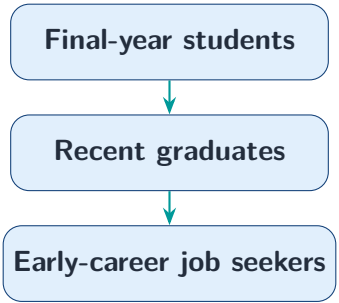
The opportunity: make career preparation clearer and more accessible.

These are the problems the prototype is designed to address; their scale and prevalence require further validation.

Who Are We Building For?

Our target users

- South African university graduates.
- Final-year students preparing to enter the job market.
- Early-career job seekers looking for entry-level roles.



User need

A straightforward way to explore relevant roles, compare skills against job requirements, and improve CVs.

Our Solution: SkillBridge AI

A graduate-focused platform combining job matching, CV analysis, and learning recommendations.

Skills-based job matching

Users select their career interests and skills to receive matches against illustrative job profiles.

Learning recommendations

Users receive suggestions for developing skills relevant to their chosen career direction.

ATS-style CV checker

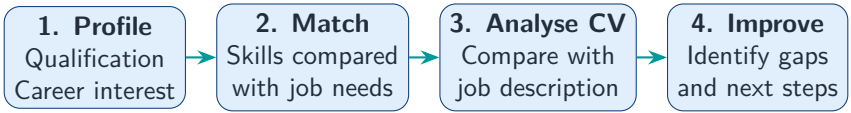
Users compare their CV text with a job description to identify matching and missing keywords.

Career preparation

The features bring several early-stage job-search tasks together in one accessible interface.

From uncertainty to practical next steps.

How SkillBridge AI Works



What the user receives

- Illustrative job matches with skills-match scores.
- A preliminary CV keyword comparison.
- Suggestions for CV improvement and skill development.

The current prototype uses rule-based matching and keyword analysis. Scores are indicative and do not guarantee recruitment or ATS success.

The Product: A Working Prototype

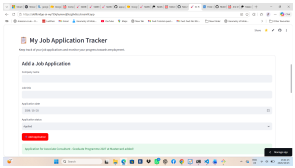
Feature 1: Job matching

- Enter a graduate profile.
- Select existing skills.
- View illustrative role matches.
- Review learning suggestions.

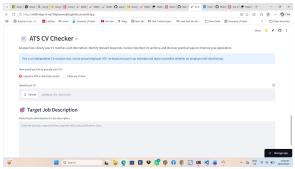
Feature 2: CV analysis

- Upload a PDF or DOCX CV.
- Enter a target job description.
- Review matched and missing keywords.
- Read preliminary suggestions.

Job-Matching Results



ATS CV Analysis Results



Screenshots from the working prototype.

Technology and Implementation

Technology	Role in the prototype
Python	Application logic and skills-matching rules.
Streamlit	Interactive web application interface.
pandas	Handling structured job profiles and analysis results.
PyMuPDF	Extracting text from supported PDF documents.
python-docx	Extracting text from DOCX documents.
GitHub	Source-code hosting and version control.
Streamlit Community Cloud	Potential hosting platform for the public demo.

Current implementation

The prototype uses transparent, rule-based skills matching and keyword comparison. More advanced AI integration is a potential future enhancement, not a claimed current capability.

Lean Business Canvas

Problem

Relevant entry-level roles are hard to find; skills gaps and CV tailoring can be unclear.

Customer Segments

South African graduates, final-year students, and early-career job seekers.

Revenue Streams

Potential university/employer partnerships and optional premium services.

Solution

Skills-based role matching, CV keyword analysis, and learning suggestions.

Unique Value Proposition

Job matching and preliminary CV analysis in one accessible platform.

Cost Structure

Development, hosting, maintenance, and future enhancements.

Key Metrics

Active users; CV analyses; matches generated; repeat usage.

Channels

Universities, career centres, graduate programmes, and student networks.

Unfair Advantage

Graduate-focused workflow for South Africa; this advantage still needs validation.

The business model is a set of hypotheses requiring user research and validation.

Potential Impact and Future Development

Intended impact

- Help graduates explore relevant career options.
- Make basic CV keyword analysis more accessible.
- Help users identify skills they may need to develop.
- Encourage structured career preparation.

Next development priorities

1. Integrate verified, current job listings.
2. Improve matching of multiword skills and phrases.
3. Add secure accounts and persistent application tracking.
4. Expand learning resources for identified skills gaps.
5. Evaluate usefulness through graduate user testing.

Measure success through user value, not just feature count.

These are intended benefits and future plans; employment outcomes have not yet been established.

SkillBridge AI

Helping graduates move from uncertainty to action.

Connect skills. Explore opportunities.
Improve your CV. Plan your next step.

Thank You

Malose Jim Mokonyama

PROJECT RESOURCES Click a button below to explore the project.

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